

year basis? Those with fixed budgets may be willing to accept lower average spending in order to create more stability for annual expenditures.

Variable spending rules are rarely evaluated with matching data and assumptions, making it difficult to adequately compare the downside risks for different strategies. If one rule suggests a 6 percent withdrawal rate while another suggests a 3 percent withdrawal rate, it doesn't necessarily mean the first rule is twice as powerful. The first rule might simply be based on higher market return assumptions. In order to properly compare different strategies, we must begin with the same set of capital market assumptions.

We should also look at where we are situated in the distribution of possible spending and wealth outcomes. Building a retirement income strategy involves numerous trade-offs, and one metric (such as the probability of failure) cannot summarize the overall performance of a strategy. Higher spending upfront means running a greater risk of having less to spend later. More aggressive asset allocation means greater upside potential for spending growth and legacy, but downside risk would increase as well. Naturally, greater spending means fewer assets for a legacy. Spending evolves differently depending on the random sequence of market returns, and retirees have to decide where to focus their concerns.

Failure Rates (or Success Rates)

The traditional failure rate (which calculates the probability of portfolio depletion) often employed by safe withdrawal rate studies cannot be used to compare variable spending strategies, as it only tracks portfolio depletion. Different variable strategies may imply different spending levels just prior to depletion. For instance, a 6 percent variable spending strategy may cause spending to fall to \$20,000 per year as portfolio depletion nears, while the 3 percent strategy might maintain spending at \$50,000 until depletion. Failure rates ignore this important distinction and focus solely on the depletion event. This reflects a general theme in variable withdrawal rate literature: the more retirees are willing to let their spending drop in retirement, the higher the initial spending rate they may use. Technically, some variable spending strategies will *never* fail. For instance, when spending is always calculated as a percentage of remaining assets, even a 99 percent withdrawal rate never runs out (though it may be tough to slice up that last remaining penny).